Quantum Mechanics II — Exercise Sheet 12

Survival, escape and decay

Variant: Solutions

Exercise 1 (A uniform energy band). A normalized state's energy density is $1/(2a)$ on $[E_c - a, E_c + a]$ and zero outside, with $E_c > a > 0$. (i) Find its survival amplitude and probability, the first positive zero, and its short-time expansion. (ii) Compare the expansion with its energy variance.

Solution. Integrating the spectral phase over the band gives

$$A(t) = e^{-\mathrm{i}E_c t/\hbar}\frac{\sin(at/\hbar)}{at/\hbar}, \qquad \mathcal{P}(t) = \left(\frac{\sin(at/\hbar)}{at/\hbar}\right)^2,$$

with the continuous value one at $t = 0$. The first positive zero is $\pi\hbar/a$. Since the variance of this uniform law is $a^2/3$,

$$\mathcal{P}(t) = 1 - \frac{a^2t^2}{3\hbar^2} + O(t^4) = 1 - \frac{(\Delta H)^2t^2}{\hbar^2} + O(t^4).$$

Exercise 2 (Decay without escape). On $L^2(\mathbb{R})$ let $H = Q$ and $\psi = \mathbf{1}_{[0,1]}$. (i) Compute its survival probability and the probability of remaining in $[0, 1]$ at time t. (ii) Explain why the limit of the survival probability alone does not prove spatial escape.

Solution. Multiplication evolution gives $\psi_t(x) = e^{-\mathrm{i}tx/\hbar}\mathbf{1}_{[0,1]}(x)$. Its survival amplitude is the integral of this phase over $[0, 1]$, so

$$\mathcal{P}(t) = \left(\frac{\sin(t/(2\hbar))}{t/(2\hbar)}\right)^2 \longrightarrow 0.$$

The position density is unchanged and supported in $[0, 1]$ at every time. Thus the probability of remaining in that interval is always one. Survival compares the vector with its initial direction, whereas spatial escape compares position probabilities in bounded regions.

Exercise 3 (A semibounded model with an algebraic tail). A state's energy density is $w(E) = a^{-1}e^{-E/a}\mathbf{1}_{[0,\infty)}(E)$, where $a > 0$. (i) Compute its survival amplitude, probability, mean energy and variance. (ii) Find the leading long-time decay and decide whether a constant exponential decay rate fits all times.

Solution. The Laplace integral gives

$$A(t) = \frac{1}{1 + \mathrm{i}at/\hbar}, \qquad \mathcal{P}(t) = \frac{1}{1 + a^2t^2/\hbar^2}.$$

The first two moments are a and $2a^2$, so the variance is a^2. At short times the probability is $1 - a^2t^2/\hbar^2 + O(t^4)$. At long times it has the algebraic behavior

$$\mathcal{P}(t) \sim \frac{\hbar^2}{a^2t^2}.$$

A constant exponential rate cannot agree with both this tail and the zero initial slope.

Exercise 4 (A finite mean with infinite energy variance). A state's energy density is $w(E) = 2a^2/(E+a)^3$ for $E \geq 0$, zero for $E < 0$, where $a > 0$. Check normalization and compute $\mathrm{Prob}(E > R)$ for $R \geq 0$. (i) Find its mean energy and decide whether the second energy moment is finite. Does the quadratic short-time theorem in Lecture 12 apply? (ii) Explain whether the existence of a first energy moment alone determines an exponential survival law.

Solution. Integrating the density from R to infinity gives

$$\mathrm{Prob}(E > R) = \frac{a^2}{(R+a)^2}.$$

At $R = 0$ this is one, which checks normalization. The substitution $E = ax$ gives the mean

$$\langle H \rangle = 2a \int_0^\infty \frac{x}{(1+x)^3}\, \mathrm{d}x = a.$$

The second-moment integrand has order $2a^2/E$, so its integral diverges logarithmically. The state is outside $D(H)$ and has no finite energy variance. The quadratic short-time theorem therefore does not apply. A first moment specifies the derivative of the survival amplitude at zero, not the full Fourier transform of the energy law. It does not determine an exponential survival law.